# Numerical Methods Project 2: Canarias

## Simulating atmospheric vortex street in the Canary Islands using Shallow Water Equations

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These projects use knowledge and codes that you have developed in the course and its tutorials, now applied to specific scientific problems. In groups of 3–4, you will write a *report* for the project and provide an **executable Python code** (this should be a `.py` file, not a notebook; you are encouraged to first work with notebooks before converting your code to `.py`) to produce all your checks and plots. Your report should contain relevant background material, answers to all questions, and a statement specifying what each group member has contributed.[1] You are strongly recommended to use LaTeX (Overleaf) for a well-formatted scientific report. It is important that you describe your procedures clearly and answer every question explicitly.

Please submit your report **and** your `.py` code to Brightspace at the latest on **Sunday 14 June, 23:59**. Do not forget to save your work regularly (at least in two places)!

The background for the project is introduced in the lecture of 11 May. You need to create your project group via Brightspace the following week. After tutorials 5–6, the remaining tutorials are dedicated to project work, although questions about the regular lectures and other course materials remain welcomed. During the first week of June we would like a short meeting with your group to check your progress; please indicate in which tutorial slot this should take place. Outside these times we are also available for questions.

## 1 Project Background

When the steady winds of the North Atlantic encounter the volcanic peaks of the Canary Islands, they do not simply flow around them. Instead, a striking pattern of alternating, counter-rotating vortices can develop downstream. Under the right conditions, these vortices stretch for hundreds of kilometres and are clearly visible from satellite images.

In this project you will implement and study the *shallow-water* equations, a simplified set of fluid equations, to simulate this phenomenon. The shallow-water equations are the simplest model that captures wave propagation and large-scale atmospheric flow, and they are widely used in meteorology and oceanography.

[1] Submissions will be checked for excessive similarity between groups. Where there is circumstantial evidence of unfair means (including sharing or distributing work, copying from sources not your own, or presenting others' work as your own), we reserve the right to lower the mark for this assignment significantly.

## 2 Part 1 — The Shallow-Water Equations (mark: 2.5)

### Background

We model the atmosphere as a single fluid layer with a mean depth $H_0$, a free-surface height $h(x, y, t)$, and depth-averaged velocities $\mathbf{u}$. $\mathbf{u}$ has two components: $u$ (east–weat) and $v$ (north–south). This model is two-dimensional (i.e., no explicit vertical component). The *shallow-water equations* (SWE) governing this 2D layer are:

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0, \tag{1}$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -g'\nabla h + \nu \nabla^2 \mathbf{u}. \tag{2}$$

Equation (1) is mass conservation. Equations (2) is Newton's second: the terms on the right are, in order, *advection* (the fluid carries its own momentum), a *pressure force* proportional to the surface slope, and *viscous diffusion* with eddy viscosity $\nu$.

The quantity $g'$ is the *reduced gravity*, an effective value that accounts for the density stratification of the atmosphere. It is expressed as:

$$g' = g\frac{\Delta\rho}{\rho}, \tag{3}$$

and taken as $g' = 0.3\,\mathrm{m\ s^{-2}}$ in this project.

**Q1)** The SWE are the simplest equations to support gravity waves. Linearize the SWE by assuming that each variable (i.e., $h$, $u$, $v$) will remain nearly constant, only departing from their average by very small variation. For instance, for $h$, this means we can write $h = \bar{h} + \delta h$, with $\delta h << \bar{h}$.

With these assumptions, show that the linearized SWE around the rest state (i.e., the mean fluid is not moving) takes the following form:

$$\frac{\partial^2(\delta h)}{\partial t^2} - g\bar{h}\nabla^2(\delta h) = 0 \tag{4}$$

**Q2)** Interpret this result. Explain why we say that the SWE supports gravity waves?

**Q3)** The presence of an island (like in the Canarias) can be is represented by an additional drag force term. Show that the full projected SWE are given by:

$$\frac{\partial h}{\partial t} = -\frac{\partial (hu)}{\partial x} - \frac{\partial (hv)}{\partial y}, \tag{5}$$

$$\frac{\partial u}{\partial t} = -u\frac{\partial u}{\partial x} - v\frac{\partial u}{\partial y} - g'\frac{\partial h}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) - C_d\,\mathcal{C}(x,y)\,u, \tag{6}$$

$$\frac{\partial v}{\partial t} = -u\frac{\partial v}{\partial x} - v\frac{\partial v}{\partial y} - g'\frac{\partial h}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) - C_d\,\mathcal{C}(x,y)\,v. \tag{7}$$

For the rest of the project, we will use a smooth Gaussian region for the drag:

$$\mathcal{C}(x,y) = \exp\left(-\frac{(x-x_c)^2 + (y-y_c)^2}{(0.35\,R_0)^2}\right), \tag{8}$$

The mountain is centred at $(x_c, y_c)$ with radius $R_0$. Where $\mathcal{C} \approx 1$ the flow is strongly slowed, mimicking a solid obstacle without complicated moving boundaries.

# 3 Part 2 — Building the solver (mark: 2.5)

## Background

To solve these equations, we will discretize on a uniform grid with spacings $\Delta x$, $\Delta y$ and advance in time with step $\Delta t$ (from initial conditions). All spatial derivatives use second-order centred differences; for example,

$$\left.\frac{\partial q}{\partial x}\right|_{i,j} \approx \frac{q_{i,j+1} - q_{i,j-1}}{2\,\Delta x},$$

where $i$ and $j$ are the indexes of the cells. The $y$-direction uses periodic boundaries (implemented with `np.roll`); the $x$-direction uses one-sided differences at the walls.

To advance in time, we will use the *second-order Runge–Kutta* method:

$$q^* = q^n + \Delta t\, F(q^n), \tag{9}$$

$$q^{n+1} = q^n + \frac{\Delta t}{2}\big[F(q^n) + F(q^*)\big], \tag{10}$$

where $F(q)$ denotes the right-hand side of the equations and $q = (h, u, v)$ is the full state. Remember: in this notation, the $^n$ exponent refers to the previous/next *step*, not a mathematical *power* operation.

**Q4)** Set up the relevant variable and initialize the the domain using the physical parameters in Table 1. The evolving variable $(h, u, v)$ should be 2D-arrays with the following initial values: $h(x, y, t = 0) = H_0$, $u(x, y, t = 0) = U_0$, $v(x, y, t = 0) = 0$. Add some very small random perturbation to $u$ and $v$ to seed the instability (i.e., this insures our flow is not starting as *perfect* but has small irregularities). Build the Gaussian mountain profile $\mathcal{C}$ (equation (8)) centred at $(L_x/3,\ L_y/2)$ and verify everything looks correct by plotting.

Table 1: Physical parameters for the Canary Islands scenario.

| Parameter | Symbol | Value |
|---|---|---|
| Domain size | $L_x \times L_y$ | $600\,\text{km} \times 240\,\text{km}$ |
| Grid resolution | $N_x \times N_y$ | $256 \times 102$ |
| Reduced gravity | $g'$ | $0.3\ \text{m}\,\text{s}^{-2}$ |
| Mean layer depth | $H_0$ | 1000 m |
| Inflow wind speed | $U_0$ | $8\ \text{m}\,\text{s}^{-1}$ |
| Island radius | $R_0$ | 20 km |
| Drag coefficient | $C_d$ | $0.02\ \text{s}^{-1}$ |
| Eddy viscosity | $\nu$ | $1000\ \text{m}^2\text{s}^{-1}$ |
| Time step | $\Delta t$ | 30 s |

**Q5)** Now, implement the three spatial operators we will need as independent function. You can call them: `ddx(q)`, `ddy(q)`, and `laplacian(q)`, where $q$ is a 2D field (it will be $h$, $u$, or $v$ when solving the equations). At the boundaries of the domain, these operators need special treatment. For the y direction, we can use periodic boundaries (i.e., use `np.roll` for instance). For the x direction, you should use one-sided differences at $x = 0$ and $x = L_x$. Verify your implementation of `ddx` by applying it to $q = \sin(2\pi x/L_x)$ and comparing with an analytical derivative.

**Q6)** Create a new function `rhs(h, u, v)` that uses your `ddx`, `ddy`, and `laplacian` functions, and returns the three time derivatives from equations (1) and (2).

**Q7)** Implement the RK2 time loop. After each update, apply the boundary conditions:
- at the **west** (inflow) boundary, impose the inflow values $u(x = 0, y, t) = U_0$, $v(x = 0, y, t) = 0$, and $h(x = 0, y, t) = H_0$;
- at the **east** (outflow) boundary, copy the second-to-last column.

**Q8)** Run for 60 000 steps ($\approx$ 20 days of model time) and plot the *vorticity* $\zeta = \partial v/\partial x - \partial u/\partial y$ every few hundred steps. Explain what the vorticity is and describe qualitatively what you observe as the simulation evolves.

## 4 Part 3 — Vortex Study (mark: 2.5)

### Background

Whether the vortex instability forms depends on two dimensionless numbers: the Froude and the Reynolds numbers. The Froude number $\mathrm{Fr} = U_0/c_0$ (where $c_0 = \sqrt{g' H_0}$ is the gravity-wave speed) compares the flow speed to wave propagation. The Reynolds number $\mathrm{Re} = U_0 R_0/\nu$ measures the ratio of inertial to viscous forces.

For a solid circular obstacle, a robust vortex street can be created if $\mathrm{Fr} < 1$ and $\mathrm{Re} \gtrsim 100$. In addition, the frequency $f$ of vortex shedding follows the empirical Strouhal relation:

$$\mathrm{St} = f\,(2R_0)/U_0 \approx 0.2. \tag{11}$$

**Q9)** Using Table 1, compute $c_0$, Fr, and Re. From the Strouhal relation, estimate the expected vortex shedding period $T_s$ in hours. Discuss whether the parameters place the flow in the vortex-street regime.

**Q10)** Record the time series of the cross-stream velocity $v$ at the wake point $(x^*, y^*) = (x_c + 3R_0,\ y_c + R_0)$. After discarding the first few days where the flow gets established, compute the power spectrum using `numpy.fft.rfft` annd `numpy.fft.rfftfreq` (read the documentation to learn how to use them).

**Q11)** From the power spectrum (or visually) identify the dominant shedding frequency and compute an empirical Strouhal number $\mathrm{St}'$ in our experiment. Compare with the expected theoretical value of 0.2. Explain your result and the differences you observed with the theory.

**Q12)** Re-run the calculation with $\nu = 5000\,\mathrm{m^2 s^{-1}}$ and compare with the original run. Compute Re in each case. Describe and explain the differences.

## 5 Part 4 — Open Section: Satellite fitting (mark: 2.5)

Go on NASA Worldview (`https://worldview.earthdata.nasa.gov`). Find a clear-sky image showing the vortex street, measure the vortex spacing, and tune $R_0$, $U_0$ and $C_d$ in your simulation to reproduce the pattern qualitatively. Discuss what physical effects are present in reality but might be absent from the shallow-water model.

Note: you might want to switch to higher order RK4 integration.